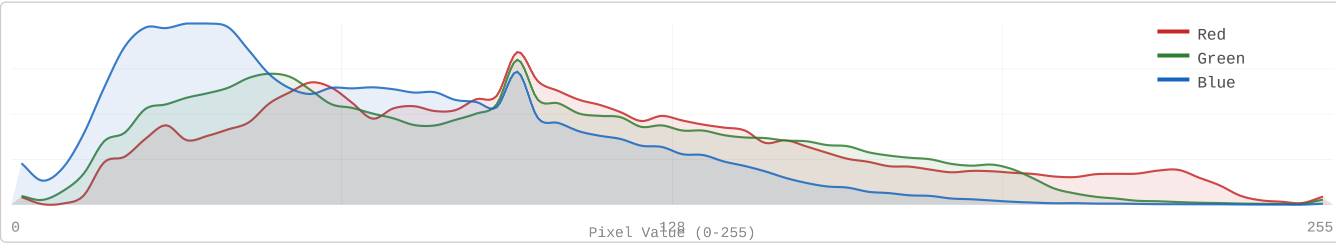
Decorrelation Gap: KODIM14

768x512 | 24-bit RGB | PNG (lossless)

Townscape with Cathedral | Two-dimensional (PC1 = 81.6%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

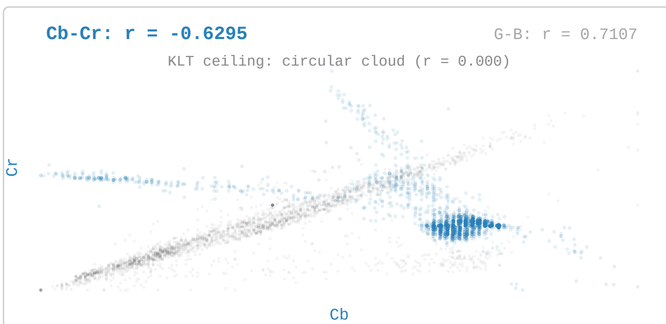
	R	G	B
R	**1.0000**	0.8613	0.4568
G	0.8613	**1.0000**	0.7107
B	0.4568	0.7107	**1.0000**

Cb-Cr:
-0.629
(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	**1.0000**	-0.5458	0.1138
Cb	-0.5458	**1.0000**	-0.6295
Cr	0.1138	-0.6295	**1.0000**

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.861318	-0.545836	0.5458
R-B → Y-Cr	0.456785	0.113824	0.1138
G-B → Cb-Cr	0.710744	-0.629452	0.6295
Avg |r|	0.676283	0.429704	0.4297

Decorrelation Gap: KODIM14

768x512 | 24-bit RGB | PNG (lossless)

Townscape with Cathedral | Two-dimensional (PC1 = 81.6%)

5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

36.5% **Low**

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

47.6%

0% Fully predictable — 100% Fully independent

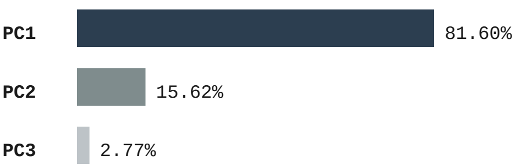
6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	39.19 dB
G	45.87 dB
B	37.95 dB
Overall	39.91 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.6571	-0.6471	-0.3868	5902.51
PC2	0.5608	-0.0768	-0.8244	1130.15
PC3	-0.5037	0.7586	-0.4133	200.52

8. PCA Derived Metrics

Condition Number

29.44

PC1/PC2: 5.22 | PC2/PC3: 5.64

Blue Channel Independence

47.6%

Variance not captured by PC1

PC1 Dominant Channel

R

Loading: R (0.66)

9. Redundancy Profile

Classification: Two-dimensional | CN: 29.44 | PC1: 81.6%

RGB Correlation: Avg |r| = 0.676283 | R-G: 0.8613 | R-B: 0.4568 | G-B: 0.7107

YCbCr Residual: Avg |r| = 0.429704 | Y-Cb: -0.5458 | Y-Cr: 0.1138 | Cb-Cr: -0.6295

Efficiency: 36.5% of KLT ceiling | Gap: 0.4297 avg |r| remaining

Blue Channel: 47.6% independent | 4:2:0 PSNR: 39.91 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.